

IDS-MID -1 SOLUTION:

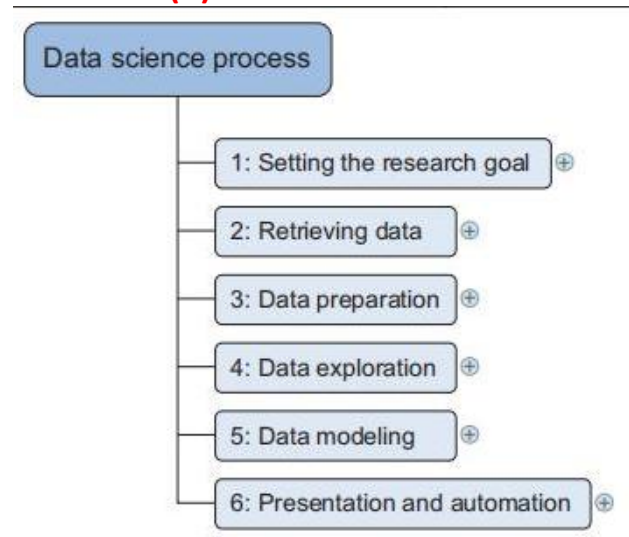
Answer#1(A):

Structured Data: Organized data stored in fixed fields, usually in rows and columns of databases. Example: A table in a relational database showing student records.

Unstructured Data: Data without a predefined format or model, difficult to store in traditional databases. Example: videos, images, emails.

Semi-structured Data: Data that does not fit neatly into tables but has some organizational tags/markers. Example: An XML or JSON document, Log files, sensor data in IoT devices.

Answer#1(B):



1. The first step of this process is setting a *research goal*. The main purpose here is making sure all the stakeholders understand the *what, how, and why* of the project. In every serious project this will result in a project charter.
2. The second phase is *data retrieval*. You want to have data available for analysis, so this step includes finding suitable data and getting access to the data from the data owner. The result is data in its raw form, which probably needs polishing and transformation before it becomes usable.
3. Now that you have the raw data, it's time to *prepare* it. This includes transforming the data from a raw form into data that's directly usable in your models. To achieve this, you'll detect and

correct different kinds of errors in the data, combine data from different data sources, and transform it. I

4. The fourth step is *data exploration*. The goal of this step is to gain a deep understanding of the data. You'll look for patterns, correlations, and deviations based on visual and descriptive techniques.
5. Machine learning or statistical models are applied to extract patterns, make predictions, or classify data.
6. The last step of the data science model is *presenting your results and automating the analysis*.

Answer#2(A):

Common Data Quality Issues

- 1) Missing Data->Some values are absent in the dataset (e.g., missing age in a customer database)
Handling Missing Data :
Remove rows/columns with too many missing values. OR Replace missing values with mean, median, mode, or predicted values.
- 2) Duplicates->The same record appears multiple times (e.g., the same customer entered twice).
Handling Duplicates:
If partial duplicates exist, merge into one complete record Or Keep first instance and delete the other fully duplicated records.
- 3) Inconsistent Data->Data does not follow a consistent format or contains errors (e.g., "USA", "U.S.A.", and "United States" stored differently, or age = 200).
Handling Inconsistent Data :Standardize and validate

Answer#2(B):

Univariate Analysis Insight

Variable: Daily sales revenue

Possible Insight: The distribution shows that weekend sales are consistently higher than weekday sales, indicating customer shopping behavior peaks on weekends.

Multivariate Analysis Insight

Variables: Advertising budget vs. Number of customers vs. Sales revenue

Possible Insight: Higher advertising budget is positively correlated with an increase in the number of customers, which in turn boosts sales

revenue. This suggests that ad spend has a direct impact on both foot traffic and revenue.

Answer#3(A):

We want a linear regression model of the form :

$Y = b_0 + b_1 X$

Where

$$\beta_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

X	Y				
1000	150	-1000	-100	100,000	1,000,000
1500	200	-500	-50	25,000	250,000
2000	250	0	0	0	0
2500	300	500	50	25,000	250,000
3000	350	1000	100	100,000	1,000,000
10000	1250	0	0	250,000	2,500,000

$$b_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{250,000}{2,500,000} = 0.1$$

$$b_0 = \bar{Y} - b_1 \bar{X} = 250 - 0.1 \times 2000 = 250 - 200 = 50$$

1. The fitted model is $y = 50 + 0.1X$.

Slope: $b_1 = 0.1$

Intercept: $b_0 = 50$

Dependent Variable: Y

Independent Variable: X

2.

3.

X	Actual Y	Predicted Y	Residual e
1000	150	150	0
1500	200	200	0
2000	250	250	0
2500	300	300	0
3000	350	350	0

4.

$$MAE = \frac{1}{n} \sum |y - \hat{y}|$$

MAE= 0/n=0

$$RMSE = \sqrt{MSE} \quad MSE = \frac{1}{n} \sum (y - \hat{y})^2$$

MSE=0/5=0 RMSE=0

$$R^2 = 1 - \frac{\text{Sum of Squared Errors (SSE)}}{\text{Total Sum of Squares (SST)}}$$

$$SSE = \sum (y - \hat{y})^2 \quad SST = \sum (y - \bar{y})^2$$

$$R^2 = 1 - \frac{0}{0} = 1$$

5. All predicted values equal the actuals, so errors are zero and $R^2 = 1$ indicates the model explains 100% of the variance in house prices — house size alone perfectly explains price in this sample. The dataset is perfectly linear (every point lies exactly on a straight line), so the fitted linear model is an exact fit.

Answer#3(B):

a) Output Variable

Linear Regression: Output variable is continuous (e.g., house price, salary, temperature).

Logistic Regression: Output variable is categorical (mostly binary: 0 or 1, yes/no, spam/not spam). Output is a probability between 0 and 1, later thresholded into a class.

b) Type of Function Used

Linear Regression: Uses a linear function:

$$\hat{Y} = b_0 + b_1 X$$

Logistic Regression: Uses a sigmoid (logistic) function to map linear combination into (0,1):

$$\hat{Y} = \frac{1}{1 + e^{-(b_0 + b_1 X)}}$$

Answer#3(C):

a) When x=0 P(x=0)≈0.881	b) When x=2 P(x=2)≈0.953
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